INT304: Network Security and Protocols – Lab 3: Network Security Protocols and Configuration

Objectives

- Understand various network security protocols and their implementations.
- Gain hands-on experience configuring network devices with security protocols.
- Learn to analyze and verify the security configurations.

Lab Activities

1. **Overview of Network Security Protocols**
    - Begin with a lecture or reading on key network security protocols such as:
        - **SSL/TLS**: Securing data in transit.
        - **IPsec**: Protecting IP packets.
        - **SSH**: Secure shell for secure remote access.
        - **VPN**: Virtual Private Network for secure connections.
        - **802.1X**: Port-based network access control.
    - Discuss the importance of each protocol in securing communications.
2. **Configuring SSL/TLS on a Web Server**
    - Set up a local web server (e.g., Apache or Nginx) on a Linux VM.
    - Obtain a self-signed SSL certificate or use a tool like Let's Encrypt for a free SSL certificate.
    - Configure the web server to support HTTPS:
        - Enable SSL/TLS.
        - Redirect HTTP traffic to HTTPS.
        - Test the configuration by accessing the web server through a web browser.
3. **Implementing IPsec VPN**
    - Set up a VPN server using software such as OpenVPN or StrongSwan.
    - Create client configurations and test connections to the VPN server.
    - Ensure secure communication between clients and the server:
        - Monitor traffic to confirm that data is encrypted.
        - Use tools like Wireshark to capture and analyze packets.
4. **Configuring SSH for Secure Remote Access**
    - Install and configure an SSH server on a Linux machine.

    - Adjust settings to enhance security:
        - Change the default port (22) to a non-standard port.
        - Disable root login.
        - Implement public key authentication instead of password-based authentication.
    - Test SSH connections from a client machine, ensuring secure access.
5. **Using 802.1X for Network Access Control**
    - Simulate a network with a switch that supports 802.1X.
    - Configure a RADIUS server to handle authentication requests.
    - Connect client devices to the network and verify that only authenticated devices can access resources.
    - Discuss potential challenges and troubleshooting methods.
6. **Verification and Testing**
    - After configuring the protocols, perform the following tests:
        - Verify HTTPS using tools like `curl` or browser security indicators.
        - Check IPsec VPN connections using ping and traceroute.
        - Validate SSH access and configuration settings.
        - Test network access with and without 802.1X authentication.

Deliverables

- Submit a report (4-6 pages) that includes:
    - Configuration details for each protocol implemented.
    - Screenshots demonstrating successful connections and security verifications.
    - A summary of challenges faced during the lab and how they were resolved.

Deadline

- The report should be submitted by the specified due date, which will be communicated in class.

Lab 4: Intrusion Detection Systems (IDS) and Traffic Analysis on Linux

Objective:

This lab aims to familiarize students with setting up and configuring an Intrusion Detection System (IDS) on a Linux machine, analyzing network traffic against the OWASP Broken Web Applications VM IP, and identifying potential security threats.

Materials Required:

- Linux operating system (Ubuntu, Kali Linux, or similar)
- OWASP Broken Web Applications VM (make sure it's running)
- Snort IDS installed on the Linux machine
- Wireshark installed on the Linux machine
- Terminal access

Lab Tasks:

1. **Install and Configure Snort:**
   - Ensure you have Snort installed. If not, install it using the following command:
   - `sudo apt-get update`
     `sudo apt-get install snort`
   - During installation, you will be prompted to set the network interface. Choose the interface that is connected to the network (e.g., `eth0`, `wlan0`).
2. **Configure Snort:**
   - Open the Snort configuration file for editing:

     `sudo nano /etc/snort/snort.conf`
   - Set the `HOME_NET` variable to the OWASP VM IP address (e.g., `192.168.56.101`).
   - Uncomment and adjust any relevant rules to ensure they are active.
3. **Start Snort in IDS Mode:**
   - Run Snort in IDS mode to monitor the network:

     `sudo snort -A console -c /etc/snort/snort.conf -i <interface>`
   - Replace `<interface>` with the actual network interface (e.g., `eth0`).
4. **Generate Network Traffic:**
   - Use a tool like `hping3` to generate malicious traffic. First, install `hping3` if not already installed:

```
sudo apt-get install hping3
```

- Generate a SYN flood attack towards the OWASP VM:

```
hping3 -S -p 80 192.168.56.101
```

- This simulates an attack on the web application running on the OWASP VM.

5. **Monitor Traffic with Wireshark:**
   - Open Wireshark and select the same interface Snort is monitoring.
   - Set a filter to display only TCP traffic:
   - `tcp`
   - Observe the incoming packets and look for SYN flood patterns.
6. **Analyze Snort Alerts:**
   - Review the Snort console output for alerts triggered by the SYN flood attack.
   - Take note of the types of alerts and corresponding packet details.
7. **Document Findings:**
   - Create a report summarizing:
     - The steps taken to configure and run Snort.
     - Types of traffic generated and the corresponding alerts triggered by Snort.
     - Screenshots from Wireshark showing packet captures during the attack.
     - Analysis of Snort's effectiveness in detecting and alerting on the simulated attack.

Submission Requirements:

- **Format:** Submit your completed report as a PDF document.
- **Content:** Include relevant screenshots, configurations, and analysis.
- **Deadline for submission:** [Insert Deadline Date]

Conclusion:

This lab provides hands-on experience in setting up an IDS on a Linux environment, demonstrating how to generate malicious traffic against the OWASP BWA VM and analyze the effectiveness of intrusion detection in a practical setting. Understanding these concepts is crucial for maintaining network security.